

ECOLOGY 404B: Integrating AI & Science Gateways into Ecology in a Digital Age

Dr. Mary K. Beals • Southern University and A&M College — Ecology 404B — FacultyHack@Gateways 2026

Project GitHub Repo: <https://github.com/DrMBea/Bayou-Ecology-CURE-2026>

Abstract

This project redesigns **ECOL 404B: Ecology in a Digital Age**, a Face-2-Face upper-level biology course at Southern University and A&M College. The redesign integrates **NAIRR CloudBank Classroom**, guided **Jupyter notebooks**, **Cornell Virtual Workshop**, and **HPC Carpentry** into a beginner-friendly course-based undergraduate research experience (**CURE**) focused on Louisiana ecology.

Students investigate ecological questions related to wetlands, biodiversity, climate change, water quality, and coastal change using browser-based tools that lower technical barriers while strengthening data literacy. The redesign preserves the course's creative, student-centered identity--virtual fieldwork, AI-supported inquiry, and science communication--while adding a clearer workflow for data exploration, visualization, interpretation, and final presentation.

The central lesson is that Science Gateway adoption in biology courses works best when the tools are accessible, scaffolded, and clearly tied to ecological reasoning.

Introduction

ECOL 404B is an upper-level Ecology course for biology majors at Southern University and A&M College. The Fall 2026 offering is Face-2-Face and organized around the theme **"Where Bayous Meet Bytes."**

The course already emphasizes Louisiana ecosystems, ecological reasoning, creativity, and digital learning. This redesign adds a clearer CURE structure and a beginner-friendly Science Gateway workflow so students can move from learning ecology as content to practicing ecology as inquiry.

The motivation is to build student confidence with data, cloud tools, and responsible AI use without overwhelming learners who are new to computational work.

Course Goals

By the end of this course, students will:

- Understand ecological concepts at the individual, population, community, and ecosystem levels.
- Apply ecological principles to real-world environmental challenges.
- Use AI tools, simulations, and browser-based Science Gateway workflows for ecological inquiry.
- Develop creative and responsible science communication skills.

Grade Computation

Category	Percent
Exams	10%
Labs	50%
Creativity	20%
Engagement	20%

Mentor Feedback Incorporated

- Keep the primary platform beginner-friendly and browser-based; use **NAIRR CloudBank Classroom** as the main student workspace.
- Support onboarding and reproducibility with **Cornell Virtual Workshop** and **HPC Carpentry** tutorials.
- Include low-bandwidth backup materials and structured checkpoints so students focus on ecological reasoning rather than technical barriers.

Target Course Description

Item	Description
Course Name	Ecology 404B: Ecology in a Digital Age
Course Number	404B
Department	Department of Biological Sciences
Enrollment	15 students
Prerequisites	Not specified in uploaded syllabus
Project Repo	https://github.com/DrMBea/Bayou-Ecology-CURE-2026

Course Description

Explore Louisiana's ecological systems through actual fieldwork, AI tools, and real-world problem solving. The course emphasizes populations, communities, ecosystems, and human-environment interactions while balancing scientific principles with creativity, storytelling, and applied problem-solving.

Key Content by Week

Module	Dates
1. Bayou Beginnings: Reading the Landscape	August 17-August 30
2. Bayou Survival Mode: Heat, Water, & Habitat	August 31-September 11
3. Counting the Bayou: Populations by the Numbers	September 14-October 2
4. Bayou Battle Lines: Networks, Niches, & Species Interaction	September 28-October 16
5. Bayou in Motion: Energy, Nutrients, & Change	October 19-October 31
6. Bayou in the Clouds: Landscapes, Climate & Conservation	November 2-November 27
7. Digital Bayou Showcase: From Finding to Story	November 30-December 4

Science Gateway Resources Used

- **NAIRR CloudBank Classroom** - cloud-hosted Jupyter environment; primary student-facing workspace for storage, browser-based access, and guided notebook activities.
- **Jupyter Notebooks / JupyterLab** - interactive notebooks for data exploration, visualization, and interpretation.
- **Cornell Virtual Workshop** - self-paced onboarding tutorials for cloud workflows and computing concepts.
- **HPC Carpentry** - reproducible workflow support, file organization, and computational habits.
- **Datasets** - NOAA Digital Coast, USGS Water Data, GBIF, plus instructor-created or simulated Louisiana ecology datasets.

These resources fit the course because they are browser-based, beginner-friendly, and support ecological reasoning without requiring advanced programming experience.

Final Project / Deliverable

Name of final project: Digital Bayou Showcase CURE Project

1. Form a small research team and define a Louisiana-focused ecological question.
2. Access instructor-curated ecological datasets in CloudBank.
3. Use guided Jupyter notebooks to organize, visualize, and interpret data.
4. Develop a final presentation supported by data visualizations and ecological analysis.
5. Present findings during the Digital Bayou Showcase.

Workflow

1. Team & Question
2. CloudBank Datasets
3. Jupyter Notebooks
4. Analyze & Interpret
5. Present Results

Project Goals

- Develop a beginner-friendly CURE module that guides biology majors through ecological question development, data exploration, visualization, and interpretation.
- Use Louisiana-focused ecological and environmental datasets to connect course concepts to wetlands, biodiversity, climate change, water quality, and coastal change.
- Build student confidence with computational ecology through guided CloudBank/Jupyter notebook activities, structured tutorials, and reflection checkpoints.

Expansion Ideas

- Extend the workflow to other biology courses such as General Biology or Botany.
- Add additional Louisiana datasets focused on coastal resilience, biodiversity, or water quality.
- Build a reusable mini-workshop for student onboarding to CloudBank and Jupyter notebooks.

Impact on Students

- Serves approximately 15 biology majors per offering in a fully asynchronous Ecology course.
- Provides no-cost access to cloud storage, guided notebooks, and Louisiana ecological datasets.
- Strengthens ecological reasoning, data literacy, teamwork, and confidence with digital tools.
- Supports creative science communication through final presentations and digital artifacts.

Lessons Learned

- Accessibility matters: a low-barrier, browser-based platform is the best entry point for beginning users.
- Scaffolding matters: tutorials, guided notebooks, and backup materials reduce friction.
- Learning goals must lead: ecological questions stay at the center while technology supports inquiry.

Course Redesign: Before & After

Initial Syllabus

- Strong ecological content, virtual fieldwork, and AI-supported learning were already present.
- Creative assignments and discussion-based engagement supported asynchronous learning.
- A Science Gateway workflow and explicit CURE sequence were not yet clearly specified.

Modified Syllabus

- Adds a beginner-friendly CURE centered on Louisiana datasets and team-based inquiry.
- Places CloudBank Classroom, Jupyter notebooks, Cornell Virtual Workshop, and HPC Carpentry into the course sequence.
- Uses guided data exploration, low-bandwidth backups, and a final Digital Bayou Showcase to align assessment with the redesign.

Authors & Mentors

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References

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- NAIRR CloudBank Classroom. <https://cloudbank.org/training/nairr-pilot-cloudbank-classroom>
- Cornell Virtual Workshop. <https://cvw.cac.cornell.edu>
- HPC Carpentry Lessons. <https://hpc-carpentry.org/lessons>
- NOAA Digital Coast. <https://coast.noaa.gov/digitalcoast>

Sponsors

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